

HL7 - Jupiter

2022-05-18

Introduction

- Communication protocol & message format for health care data
- Std for exchange b/n medical systems
- HL7 v2.x – 95% of Healthcare systems use it
 - HL7 v2.3.1 – 90% of HL7 v2.x is primarily used
- Pipe delimited format

Trigger Events (Relevant)

- ORM – Order Entry Message
- ORU – Observation Result Message - Unsolicited

Segments (Relevant)

- OBR – Observation Request [ORM | ORU]
- OBX – Observation Segment [ORM | ORU]
- ORC – Common Order Segment [ORM]

Key Difference between the two main types

ORM – Order Entry Message

- MSH|^~\&|MESA_OP|XYZ_HOSPITAL|iFW|ABC_RADIOLOGY|||ORM^O01|101104|P|2.3|||
- PID|1||20891312^^^^EPI||APPLESEED^JOHN^A^^MR.^|19661201|M||AfrAm|505 S. HAMILTON AVE^^MADISON^WI^53505^US^^^DN|DN|(608)123-4567|(608)123-5678|S||11480003|123-45-7890|||^^^W|^
- PD1|||FACILITY(EAST)^12345|1173^MATTHEWS^JAMES^A^^
- PV1|||^^^CARE HEALTH SYSTEMS^^^^|||1173^MATTHEWS^JAMES^A^^|610613|V
- ORC|NW|987654^EPIC|76543^EPC||Final||^^^20140418170014^^^|20140418173314|1148^PATTERSON^JAMES^^^|1173^MATTHEWS^JAMES^A^^|1133^^^222^^^|(618)222-1122|
- OBR|1|363463^EPC|1858^EPC|73610^X-RAY ANKLE 3+ VW^^^X-RAY ANKLE |||||1173^MATTHEWS^JAMES^A^^|(608)258-
- 8866|||Final||^^^20140418170014^^^|||6064^MANSFIEL D^JEREMY^^^|1148010^1A^EAST^X-RAY^^^|^
- DG1||I10|S82^ANKLE FRACTURE^I10|ANKLE FRACTURE|

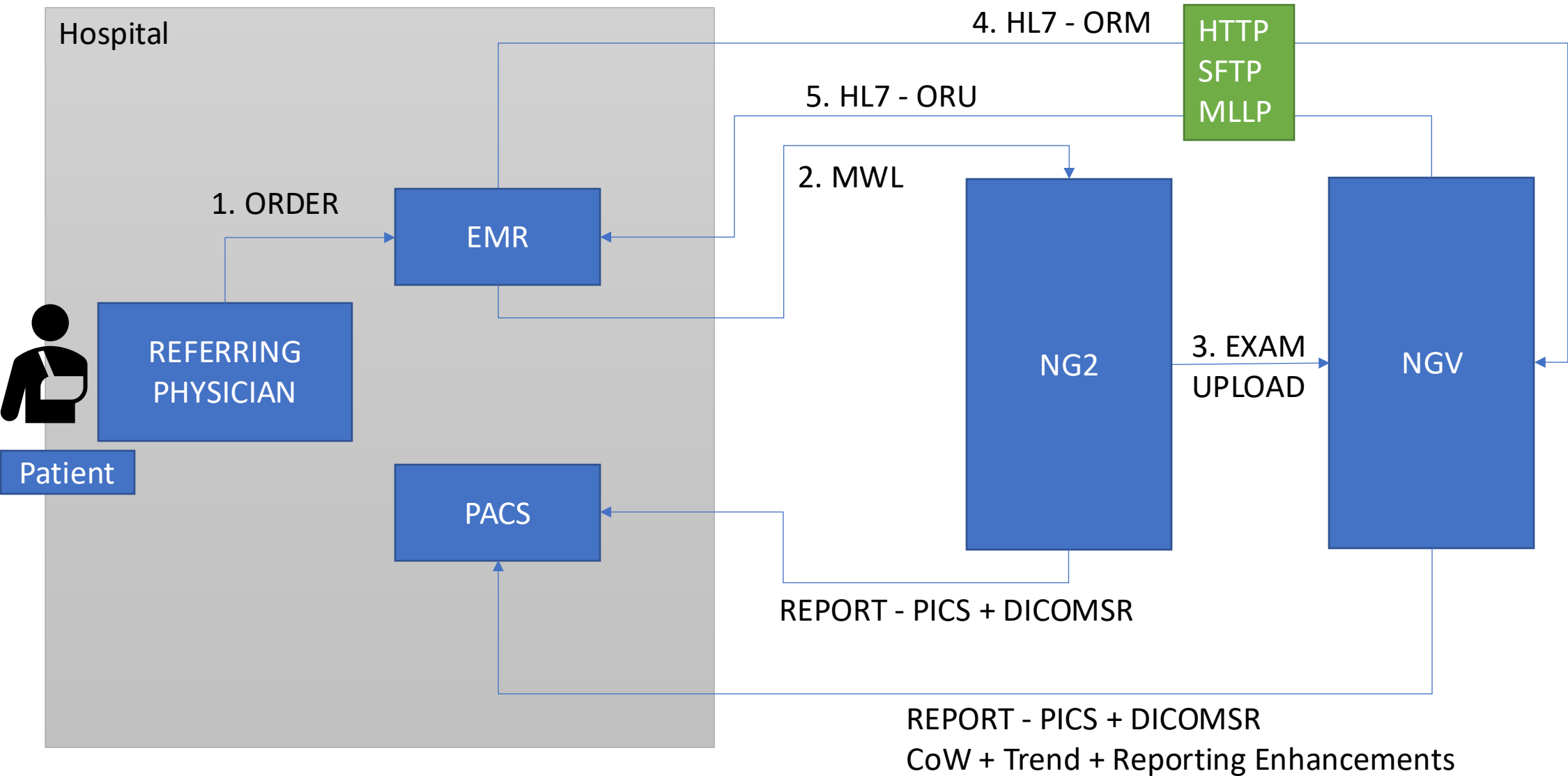
ORU – Observation Result (Unsolicited)

- MSH|^~\&|MESA_RPT_MGR|EAST_RADIOLOGY|iFW|XYZ|||ORU^R01|MESA3b|P|2.4|||
- PID|||CR3^^^ADT1||CRTHREE^PAUL|||PatientAcct|||
- PV1||1|CE|||12345^SMITH^BARON^H|||
- OBR|||20010501141500.0000|||F|||
- OBX|1|HD|SR Instance UID||1.113654.1.2001.30.2.1|||F|||
- OBX|2|TX|SR Text||Radiology Report History Cough Findings PA evaluation of the chest demonstrates the lungs to be expanded and clear. Conclusions Normal PA chest x-ray.||||F|||
- CTI|study1|^1|^10_EP1

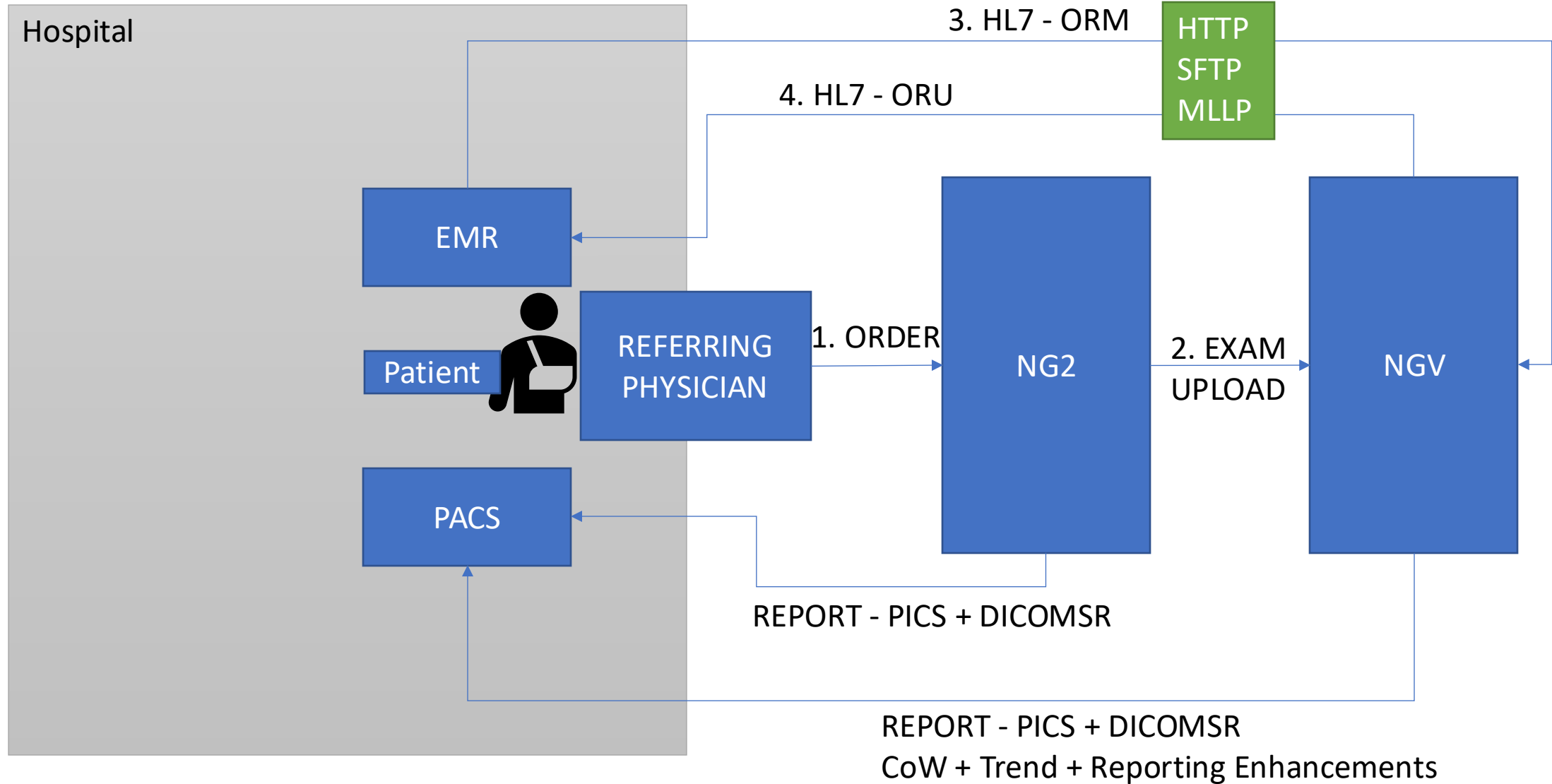
HL7 - Jupiter

- Bidirectional communication between Novasignal Platform & EMR
- Support for all major versions of HL7 - 2.1, 2.2, 2.3, 2.3.1, 2.4, 2.5, 2.5.1, 2.6, 2.7, 2.8, 2.8.1, 2.8.2
- Network support for security through VPN
- Support for various protocols – HTTPS, SFTP, MLLP
- Additional support
 - Order in the cloud
 - Partial ORU creation.

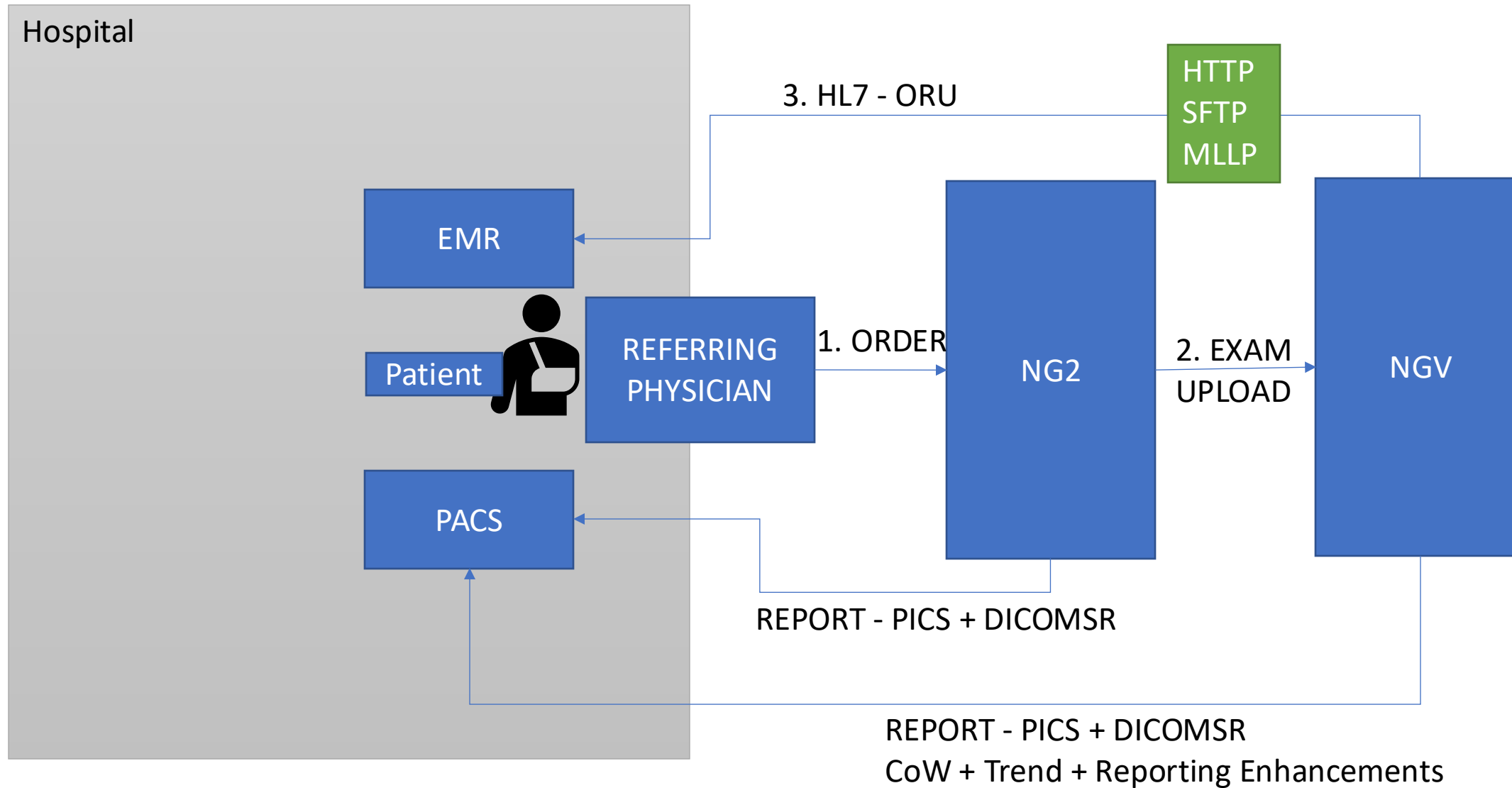
HOSPITAL – NOVASIGNAL - NORMAL

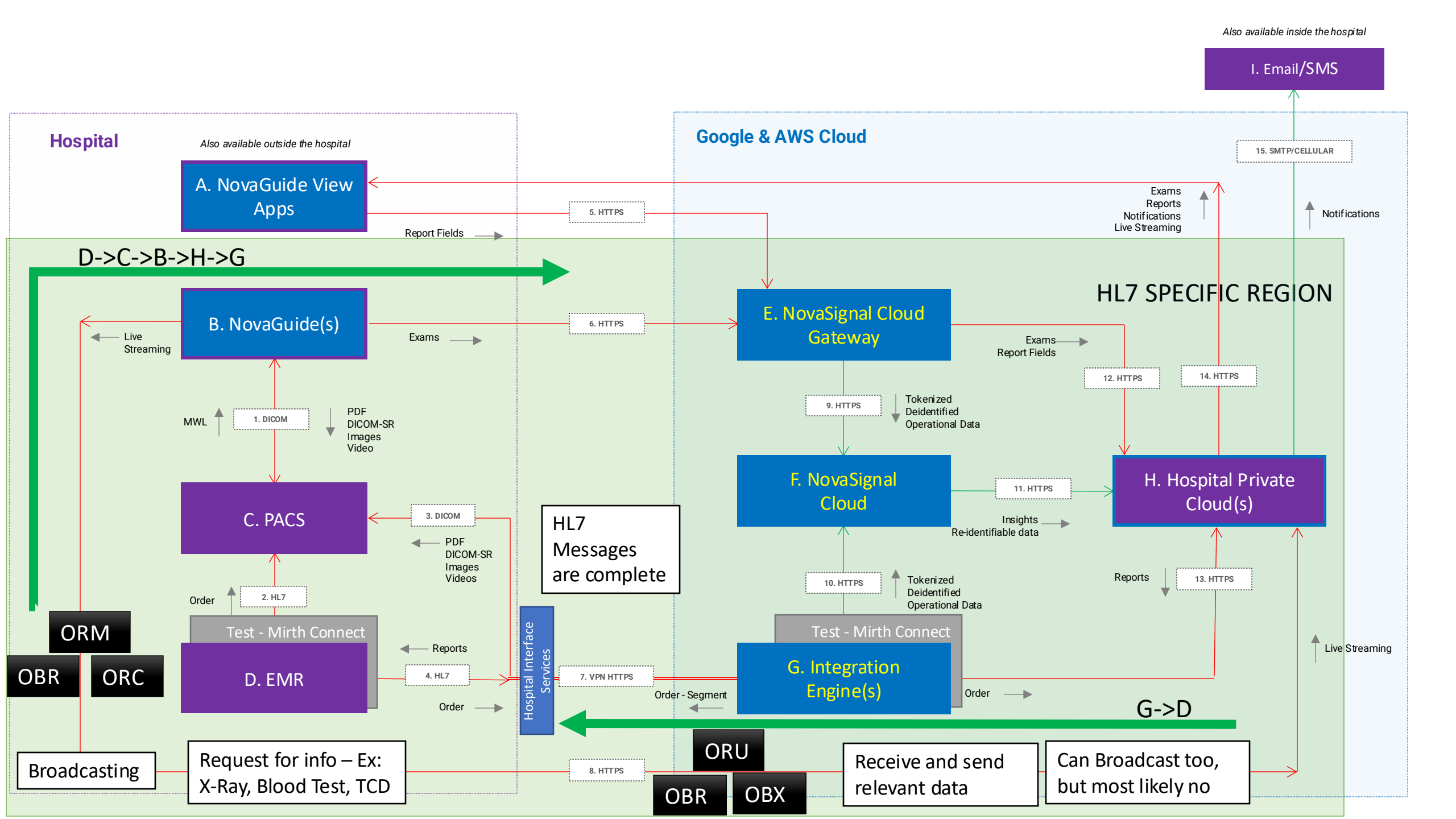


HOSPITAL – NOVASIGNAL – ORDER ID IN CLOUD



HOSPITAL – NOVASIGNAL – PARTIAL ORU

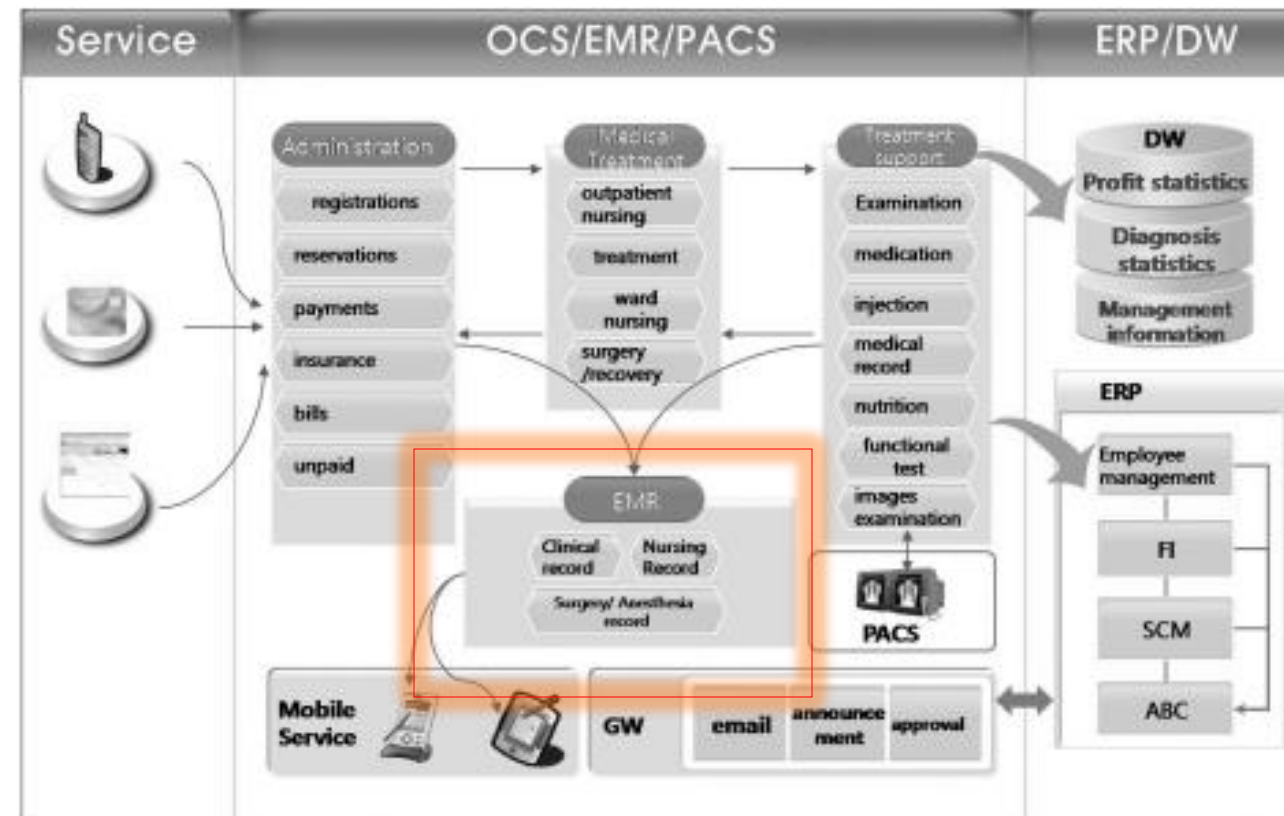




EMR Simulator

EMR (Electronic Medical Records) Simulator

- EMR - Digital version of all the information you'd typically find in a hospital's paper chart:
 - medical history, diagnoses, medications, lab results and doctor's notes.
- Simulate the hospital EMR
 - Mirth Connect – Too simplistic and very specific
 - Own implementation –scale easily and include various features – possible because of understanding medical technologies - DICOM and HL7
- Components
 - [Flask] API Server (HTTP)
 - [Pynetdicom] SCP Server (DICOM)
 - [HL7 APY] TCP Server (MLLP)
- Implementation
 - 3 Docker Containers in a Container Optimized Virtual Machine.

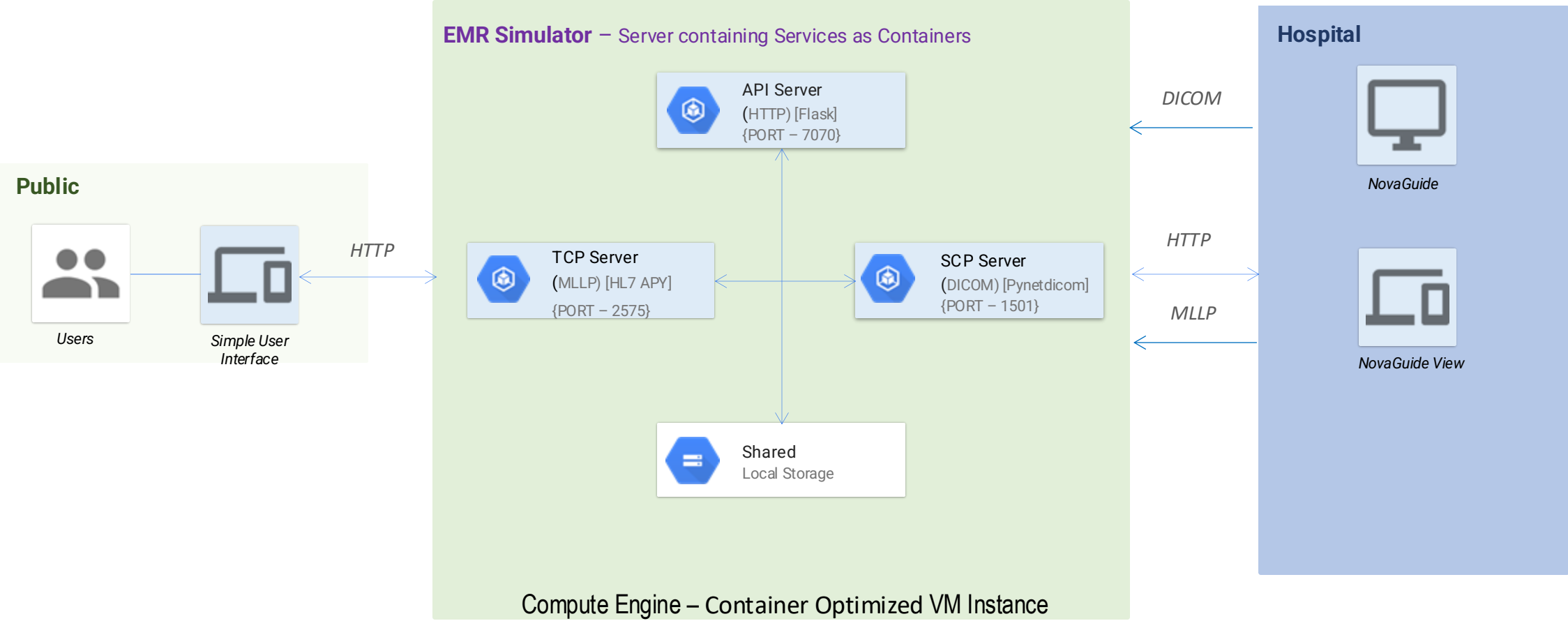


Requirements

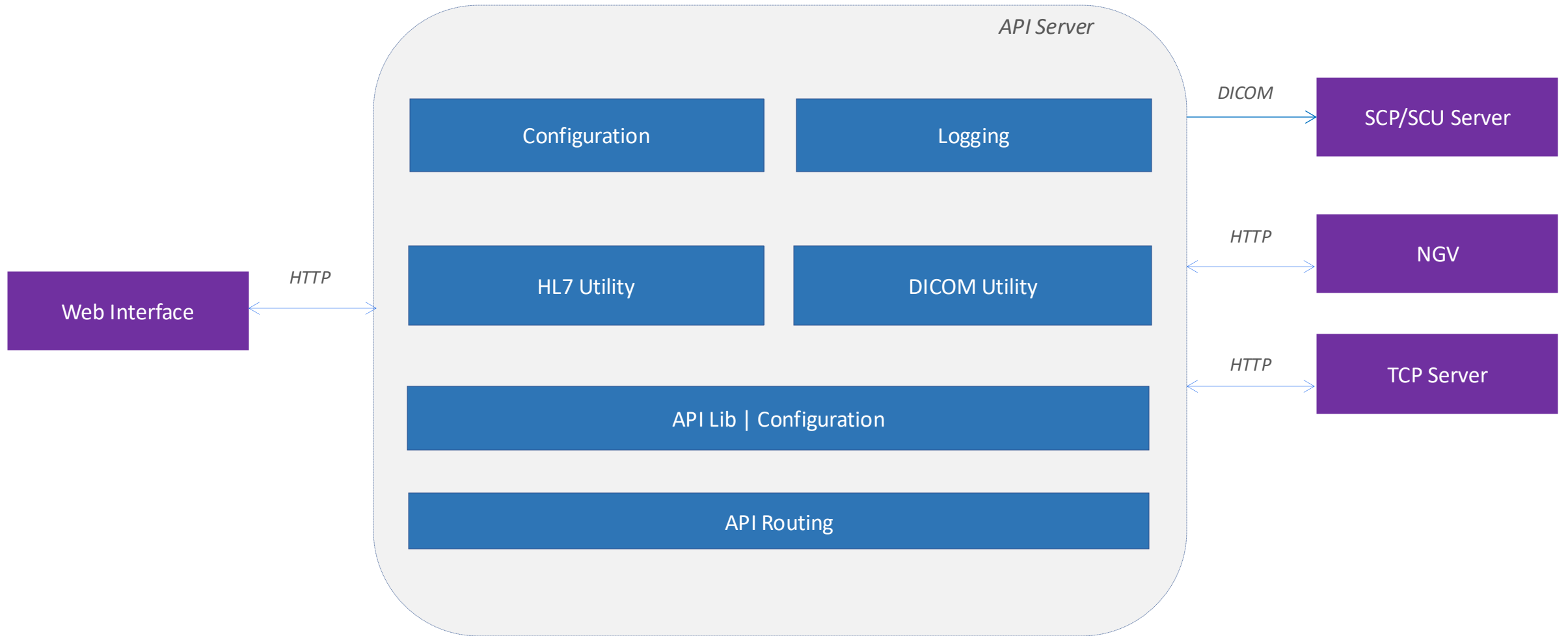
1. Create/Edit/List/Delete orders using HTTP Interface
2. DICOM SCP Server that returns MWL based on orders
3. Provide MLLP interface to submit order results through ORU Message
4. Store HL7 messages as separate files
5. View HL7 message using the browser
6. Download PDF encoded through HL7 message

Architecture

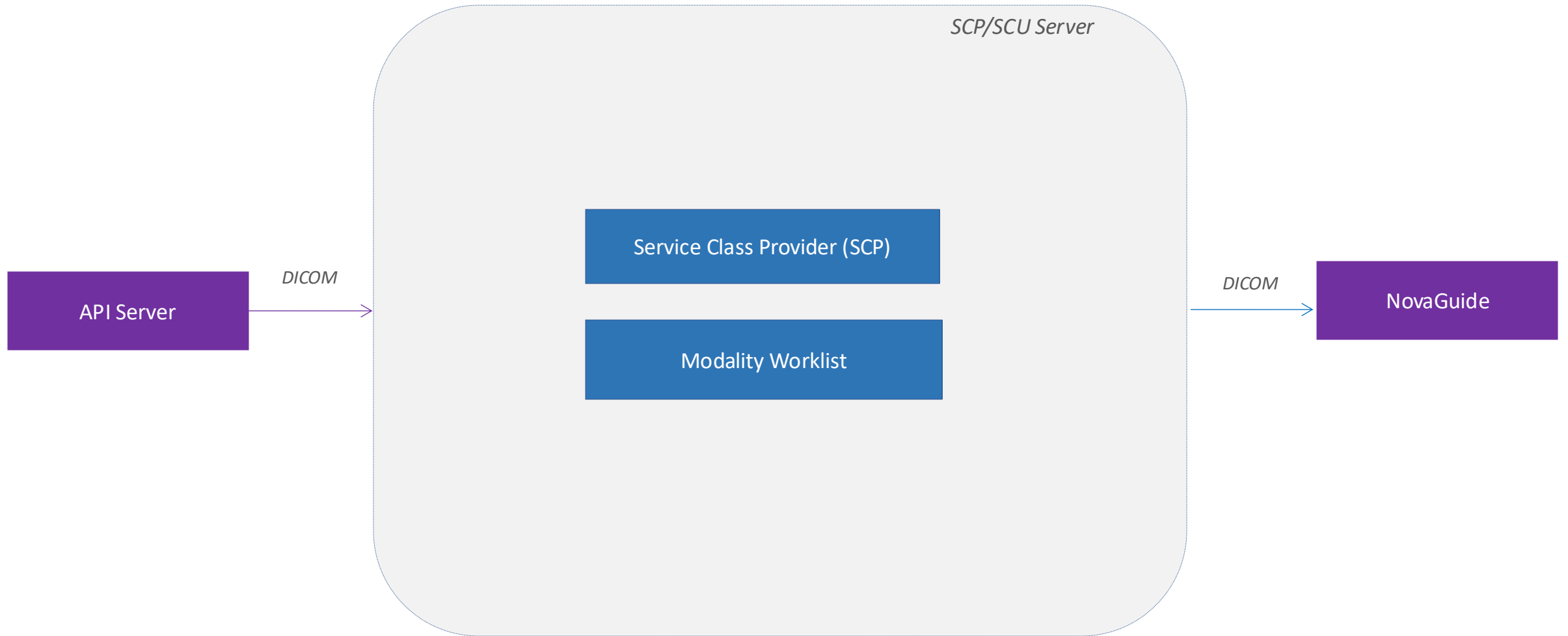
EMR Simulator Architecture –



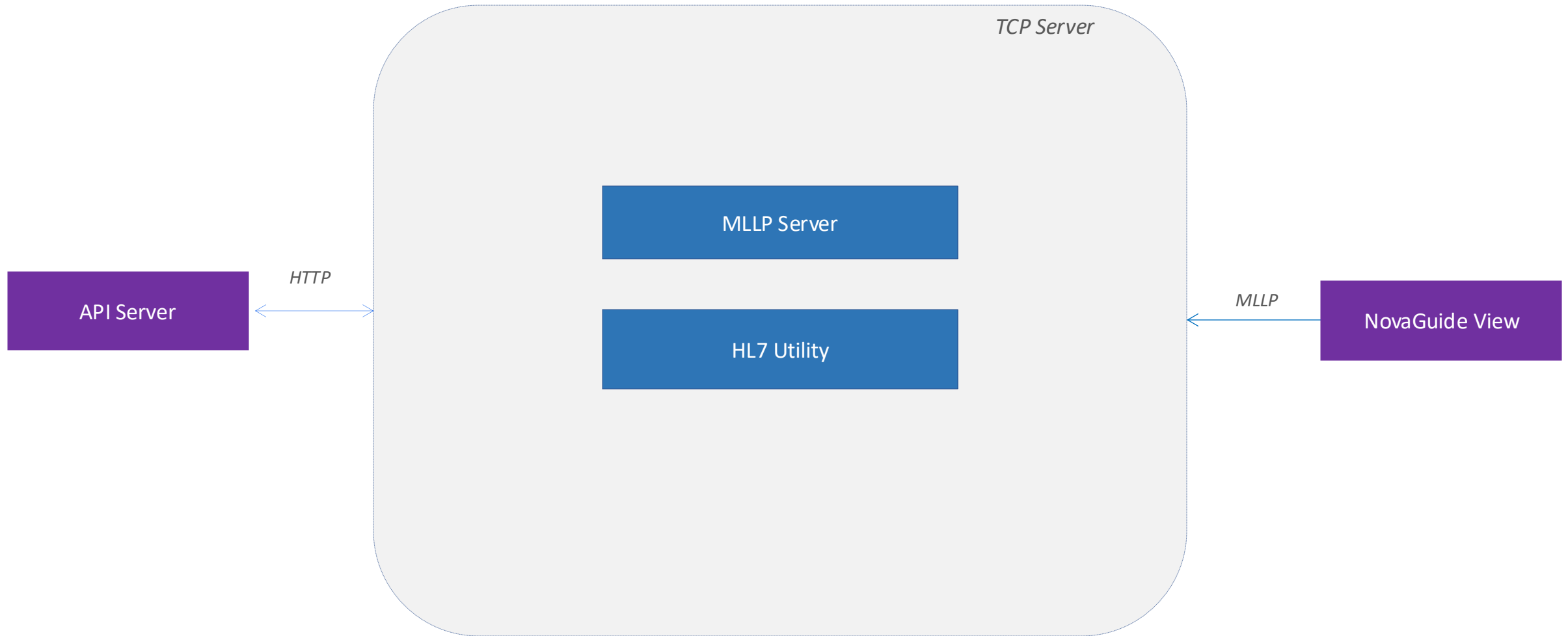
API Server - (HTTP) [Flask] {PORT – 8080}

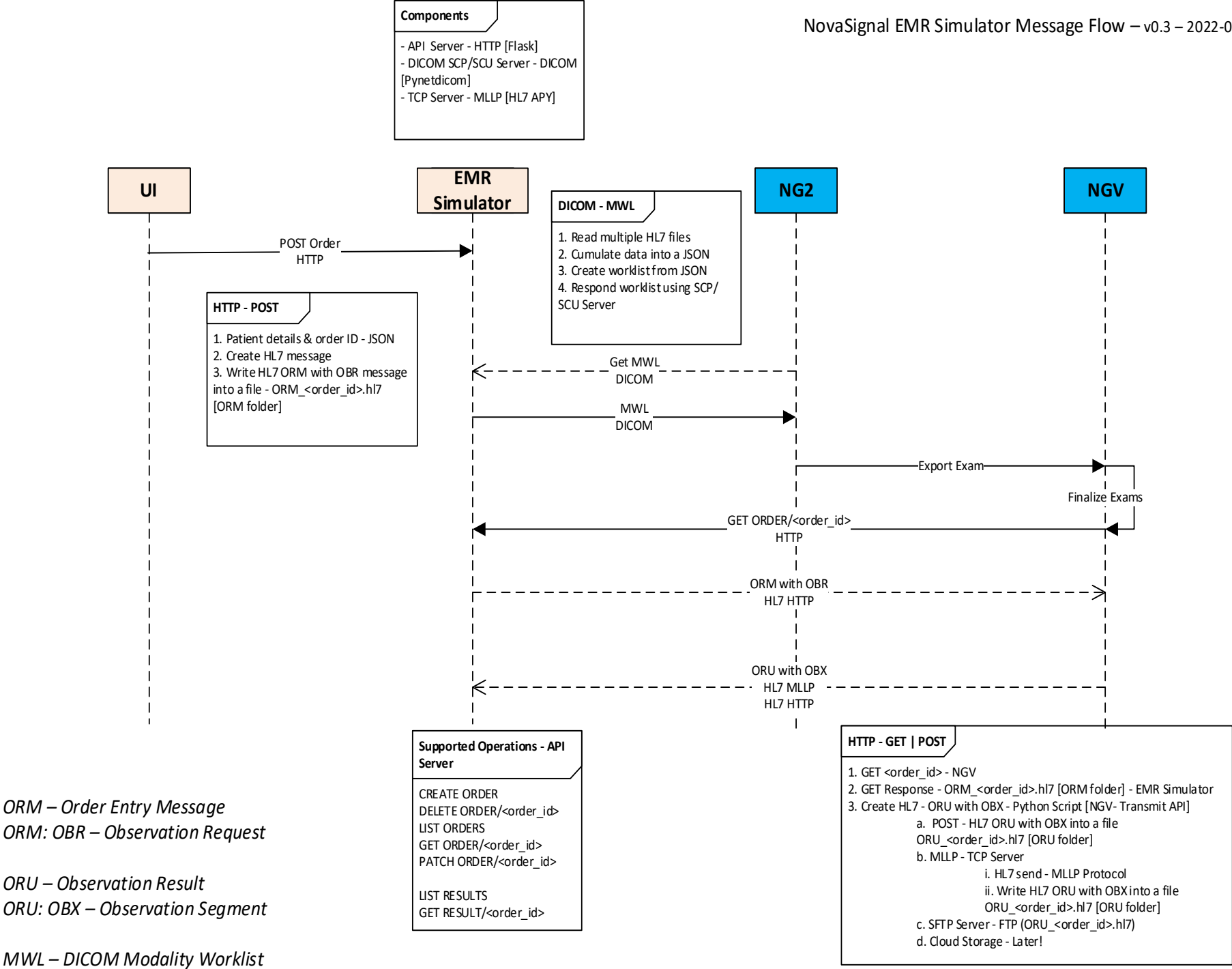


SCP Server (DICOM) [Pynetdicom] {PORT – 1234}

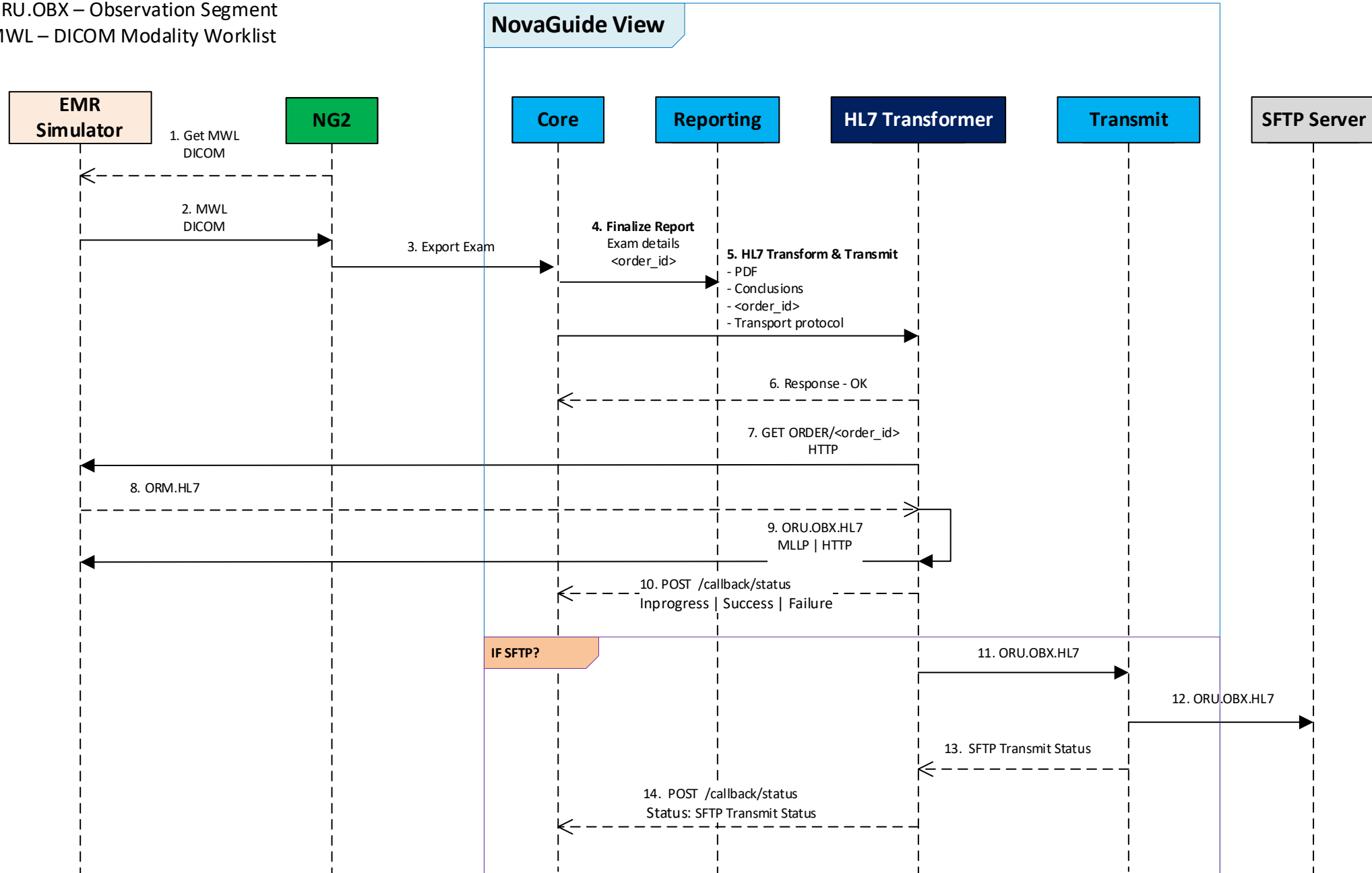


TCP Server (MLLP) [HL7 APY] {PORT – 6661}





ORM – Order Entry Message
ORM.OBR – Observation Request Segment
ORU – Observation Result
ORU.OBX – Observation Segment
MWL – DICOM Modality Worklist



Web User Interface

Home Page – HL7 Orders

Search Order

Search Create Order

Sl No	Order ID	Patient Details	Status	ORM	ORU	Edit / Delete
1	0001	Raymond Complete	Created	MSH		Edit / Delete
2	0002	John Doe	-	MSH		Edit / Delete
3	0003	Jane Doe	Fulfilled	MSH	MSH	Edit / Delete
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.

Order details

Order Details – Create | Edit

Enter the Order ID :

0001

Enter the Patient Details:

-Patient Name

-Patient ID

-Accession Number

Exam type

Notes

Submit

Save

Cancel

Put in all the details except time stamp into free form fields

Edit Order

Delete Order

VPN

Proxy

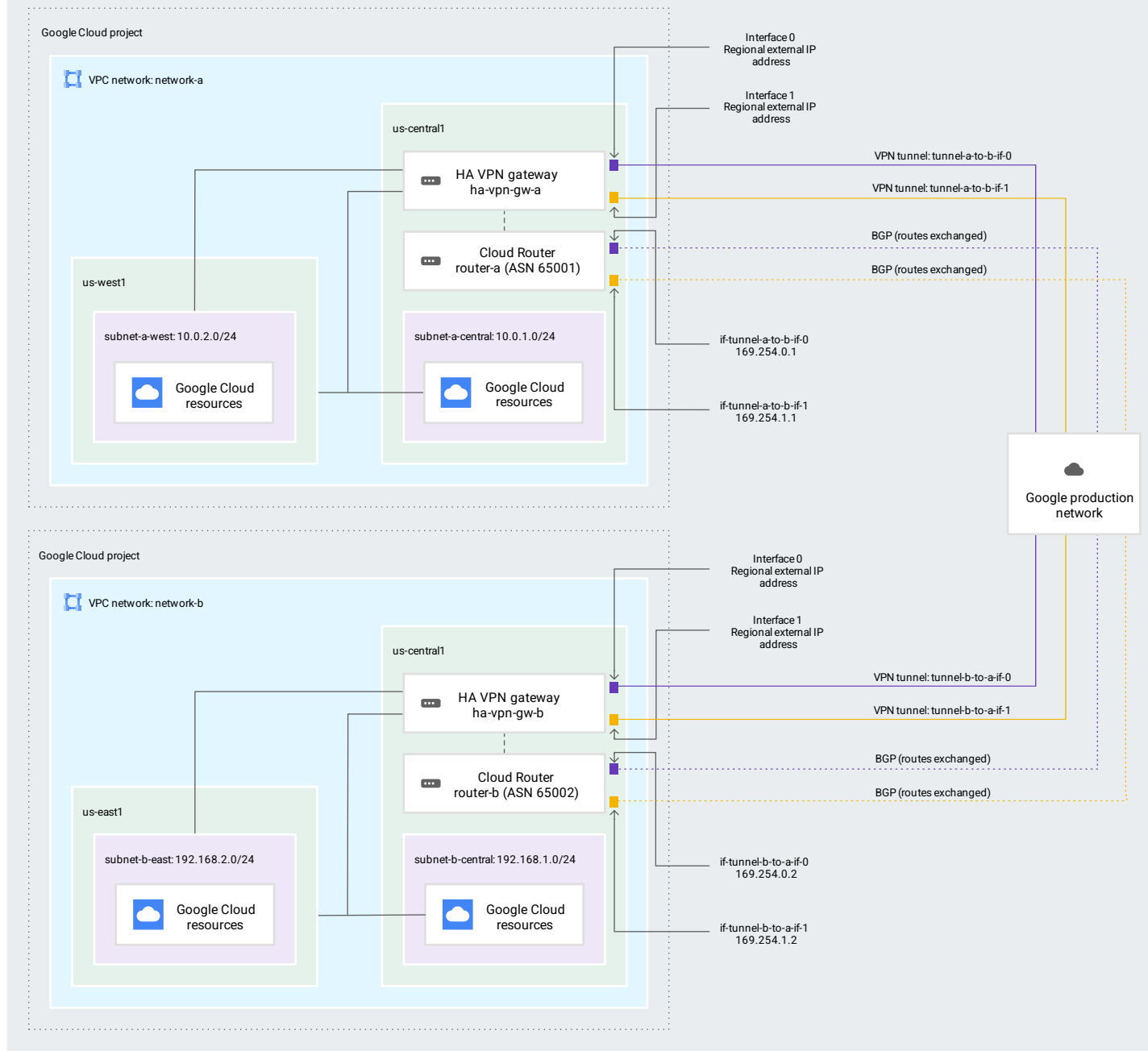
- HL7 transform API – VM to be run in container optimized – project: dev-aran
- VPN from HL7 transform VM to EMR Simulator VM
- EMR Simulator IP Address
 - Public – order – access as if in hospital
 - VPN – call from transform API – SFTP , MLLP comes from

Components

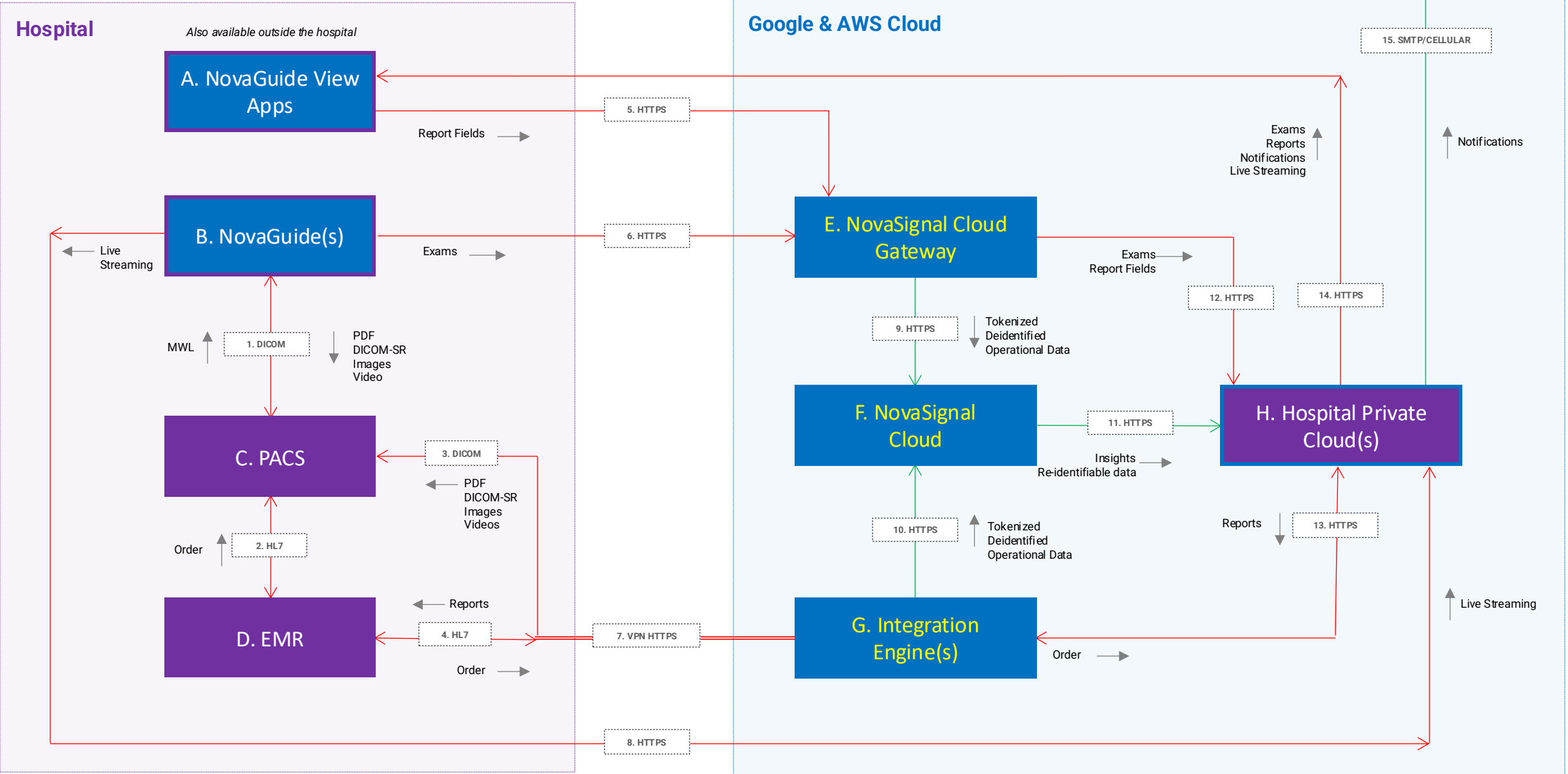
- VPC
- Subnets
- Cloud Routers
- VPN
- VPN Tunnels
- BGP Sessions

General Guidelines

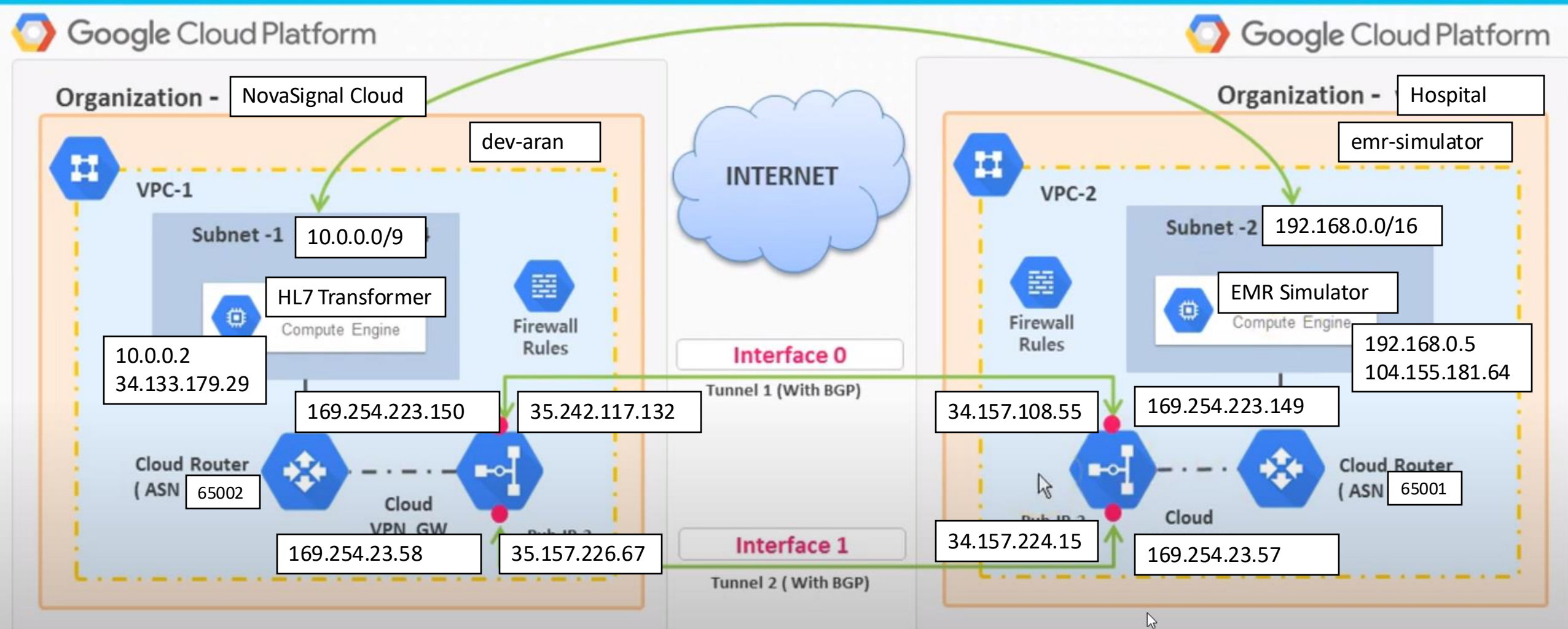
- Make a network diagram for easy troubleshooting
- Follow GCP guidelines to the T
- VPC's should be created before doing any VPN
- VPN tunnels should be connected by copying the shared key
- BGP sessions needs to be done automatically first and manually copy the IP address from one to the peer
- Check for VPN tunnels & BGP Session statuses
- Subnets for two separate projects should have different IP ranges for internal communication
- Firewall rules to be applicable to the correct subnet, the necessary ports to be opened
- The VM instances must be connected to the subnets created specifically for the VPN connections
- Ping the other IP addresses, Check the firewall rules if there is any issue with connectivity
- Block the external IP addresses soon enough



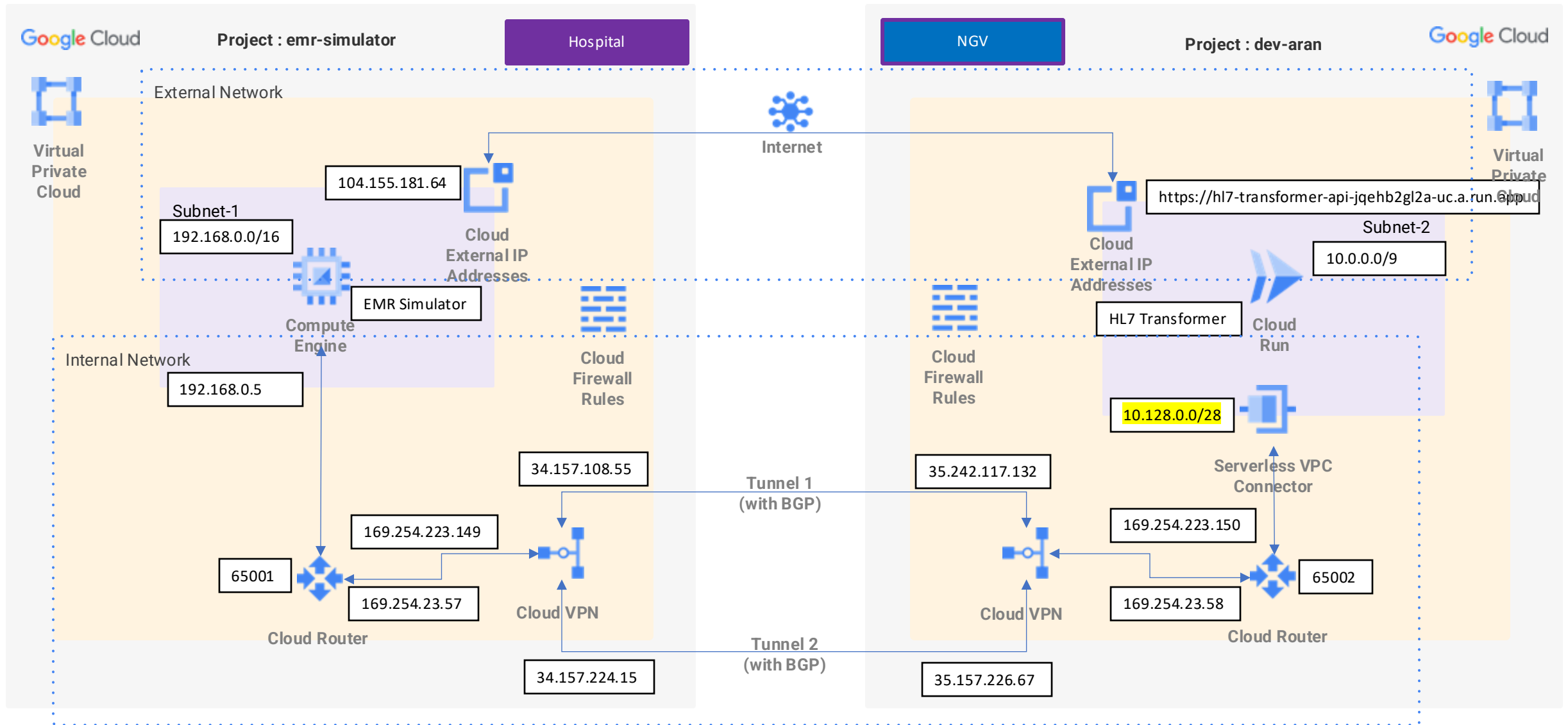
NovaSignal CJ Platform Architecture



Network Diagram



Network Diagram: Hospital - NGV



Cost Estimate

Average pdf size * no of exams – HL7 – Very little
Average exams size for DICOM-PACS – 100GB

100 GB/month = $0.11 * 100 = \$11 \rightarrow$
>>~\$100 for VPN<<

- <https://cloud.google.com/network-connectivity/docs/vpn/pricing>

Gateway	Egress traffic	Ingress traffic	IP address	Total
us-central1 gateway (\$0.050) x 2 tunnels x 720 hours = \$72.00	2 TB (or 2,048 GB) x \$0.11 = \$225.28	No charge for ingress traffic.	No charge for a reserved external IP address that is used by a tunnel.	\$297.28

Gateway	Egress traffic	Ingress traffic	IP address	Total
asia-northeast1 gateway (\$0.075), with four tunnels x 720 hours = \$216.00	Traffic to the data center: 10 TB (or 10,240 GB) x \$0.14 = \$1,433.60	No charge for ingress traffic.	No charge for a reserved external IP address that is used by a tunnel.	\$3,288.00
	Traffic to Network B: 20 TB (or 20,480 GB) x \$0.08 = \$1,638.40	No charge for ingress traffic.		
eu-west1 gateway (\$0.065) x 2 tunnels x 720 hours = \$93.60	Traffic to Network A: 20 TB (or 20,480 GB) x \$0.08 = \$1,638.40	No charge for ingress traffic.	No charge for a reserved external IP address that is used by a tunnel.	\$1,732.00
southamerica-east1 gateway (\$0.075) x 0 tunnels x 720 hours = no charge	No egress traffic.	No ingress traffic.	One unused external IP address in southamerica-east1 (\$0.015) x 720 hours = \$10.80.	\$10.80
Grand total				\$5,030.80

Tunnel cost is more but
egress traffic cost/gb is less

HL7 Transform

Message Sequence Chart

- https://neuralanalytics.sharepoint.com/:p:/r/sites/cloud/Shared%20Documents/4a_Jupiter/0_Requirements%20%26%20Design%20Inputs/Architecture/NS%20Cloud%20App%20Architecture%20-%20Jupiter.pptx?d=wfd263412834c4f00a2174d70b159b6d6&csf=1&web=1
- Slide #27
- [NS Cloud App Architecture - Jupiter.pptx](#) – Partial ORU Message
- [NS Cloud App Architecture - Jupiter.pptx](#) – Normal HL7

New ECC Check TODO:@Vivek

- HL7 - Order number is required - If not present - allow them to edit or modify?
 - Order present - edit and change
 - Order absent - create and edit - Should we prompt

HL7 Transmission – Design details

Summary Table

Status – Global Level

Total _____

Accepted _____

Transmitted _____

Failed _____

Buttons for Global level

Download

Refresh

Retransmit

Resume – Only when failed

Status for Global level

Accepted/Failed

Success

Status for file level

Accepted/Failed

Transmitting

Transmitted/Failed

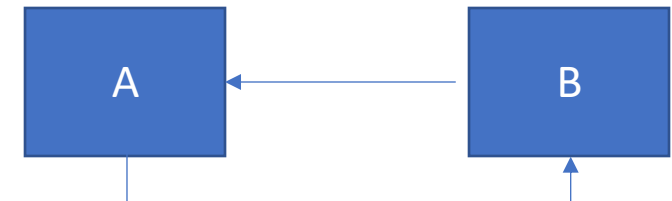
- EMR

- To MLLP –

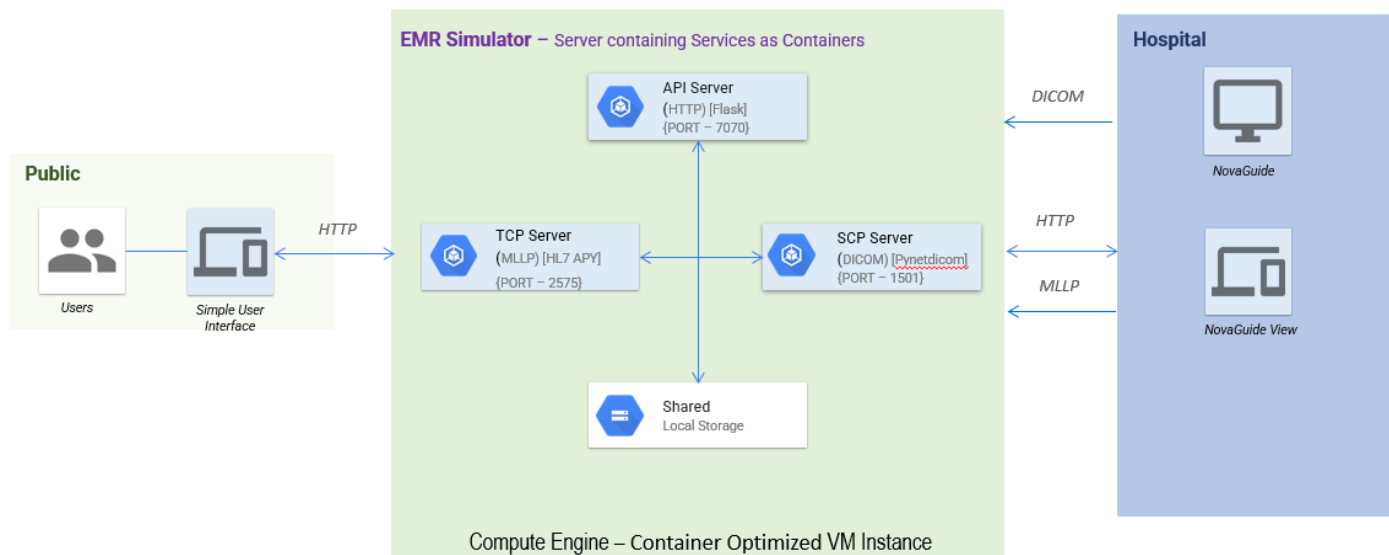
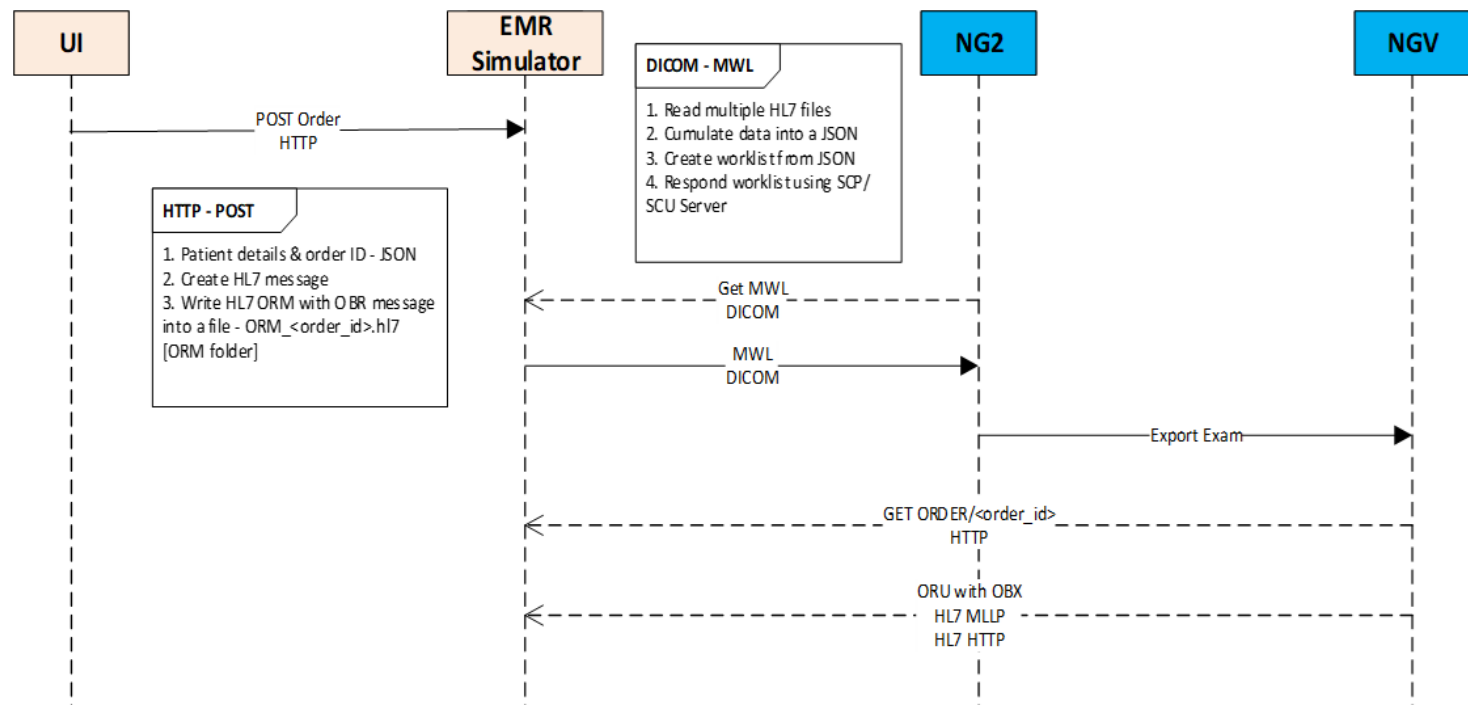
- XXXXX.ORU - TRANSMITTED

- To HTTPS –

- To SFTP -



Study



Mirth Connect - HTTP - POST

- HTTP Listener natively doesn't have a way for endpoints to have a particular method – GET | POST | PUT
- Solutions investigated
 - HL7 message created in python script and converted into base 64 and sent to Mirth Connect, written into a file in the docker (issue with querying it later)
 - Multipart message Field sent to Mirth Connect, transformed to HL7
 - XML sent to Mirth Connect, transformed to HL7
- Database saving to be discussed? Whether it is necessary or not
- Order ID – orders folder – order with filename

Mirth Connect - DICOM MWL

- DICOM SCP/SCU not natively supported in Mirth connect
- Solutions investigated
 - Patch in 2.1 version but not working well
 - Implement with Java & JavaScript in Mirth Connect – tried but a lot of help would be required since not familiar
 - Python Server – pynetdicom working as SCP/SCU – but a different implementation must be done
 - DICOM SCP – 50-00115 – Field file can be kept and consumed by the tool and can be started. Since the testing is done

HL7 Jupiter Requirements

- 5 – HL7 T&T - API Spec = Link to be sent (Model it on Reporting) – Core -> HL7 Transform – Test cases to be written – Success cases not the errors.
- Specs as code folder – Pytest code – Test Code – Specification -> API Call | 5 & 11
- 11 – Success and error
- Option to be given for the PDF to be converted to gray scale
- Grayscale/Normal PDF to be encoded in base 64 NOT the link
- Conclusions or other details to be checked which is supposed to come in the HL7 – TODO – Fields names (Conclusions, Notes, etc) – All text fields that user types not BP and other patient vitals only pure text. (Consultant would be the best person to take decision)
- ORU message specifications
- Database entry – ways to be specified – Text field in an exam details row (if at all then only the ORU) [4GB can be stored so no issues]

Marketing

HL7 - Jupiter

- High Level Pic of HL7 features (Uncluttered) - TODO@Vivek

Archive

Introduction

- Communication protocol & message format for health care data
- Std for exchange b/n medical systems
- HL7 v2.x – 95% of Healthcare systems use it
 - HL7 v2.3.1 – 90% of HL7 v2.x is primarily used
- Pipe delimited format
- Tools
- Python Libraries
 - Python-hl7 – v2.x parsing
 - HL7apy - Creation
- MLLP(Minimal Lower Level Protocol)
- MirthConnect – Channels – Write, transform, change protocol

Trigger Events (Relevant)

- ORM – Order Entry Message
- ORU – Observation Result Message - Unsolicited

Segments (Relevant)

- OBR – Observation Request [ORM | ORU]
- OBX – Observation Segment [ORM | ORU]
- ORC – Common Order Segment [ORM]

EMR Simulator – #! TODO: Model it as per user's page

Create Order ID

Enter the Order ID :

Enter the Patient Details:

Field1 – Patient Nam

Field2 – specific Exs

Field3

Submit

Cancel

Edit Order ID

Enter the Order ID :

0001

Enter the Patient Details:

Field1

Field2

Field3

Save

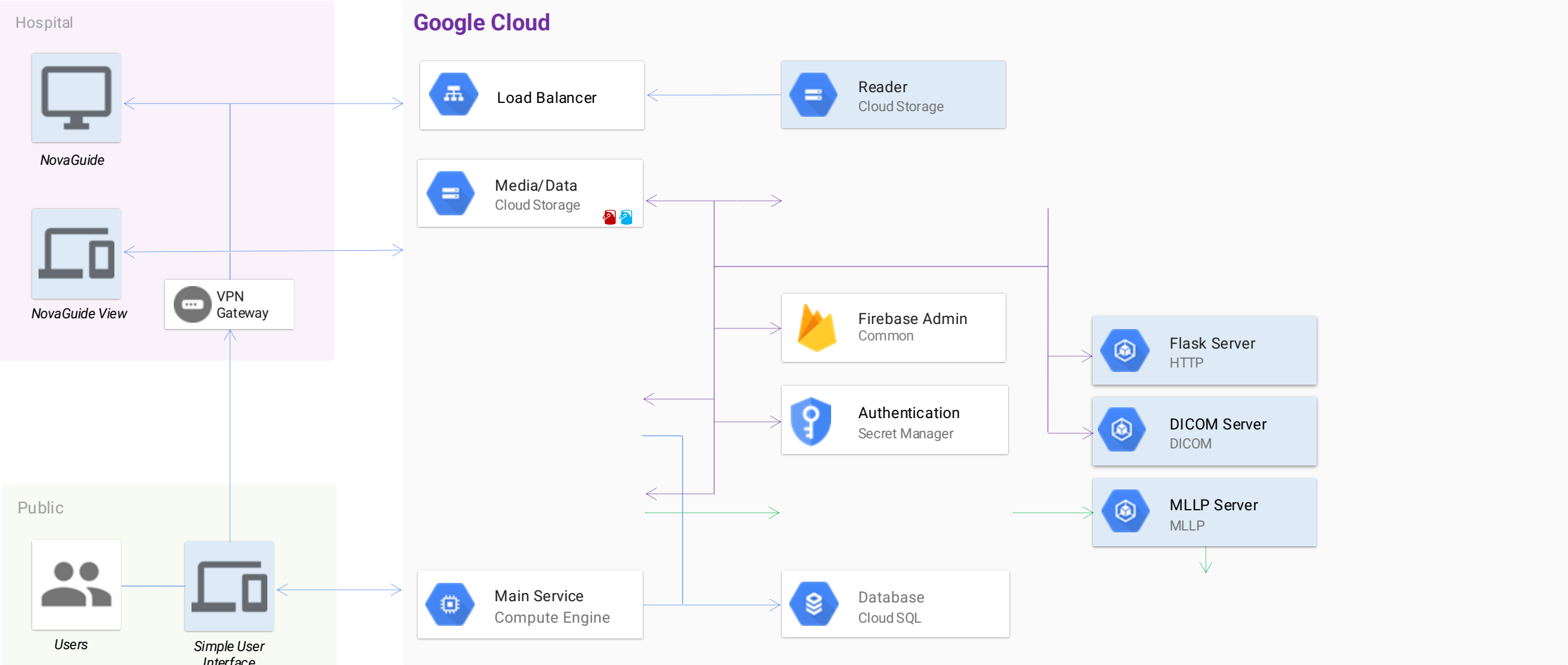
Cancel

Order Details						
SI No	Order ID	Patient Details	Status	Details		Edit / Delete
				ORM	ORU	
1	0001	New	Created	MSH		Edit / Delete
2	0002	New1	-	MSH		Edit / Delete
3	0003	New2	Received	MSH	MSH	Edit / Delete

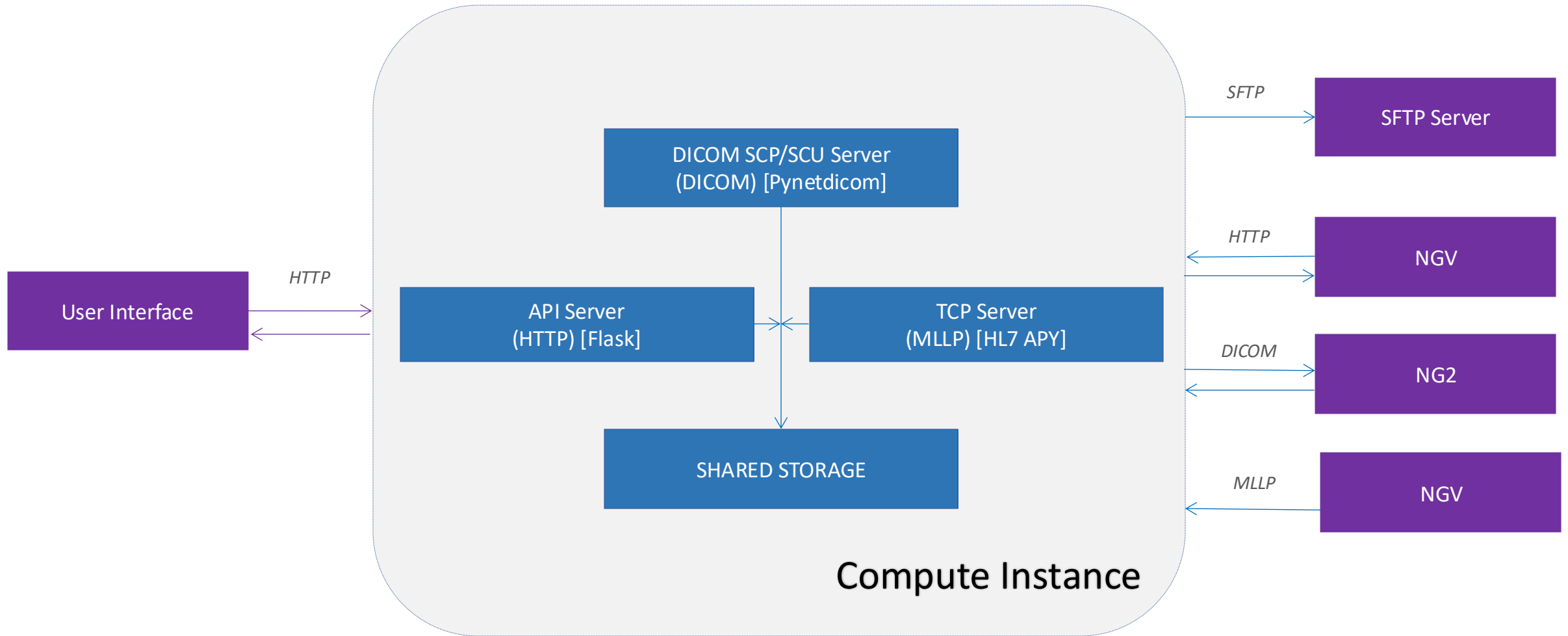
Components

- API Server (HTTP) [Flask]
- DICOM SCP Server (DICOM) [Pynetdicom]
- TCP Server (MLLP) [HL7 APY]
- Key Features
 - Docker Compose since different docker containers are created for each of these services Assign a port number for each of these services since all will use same IP
 - Reason Compute Engine(VM Instance) not Cloud Run(HTTP only) – All three-service using different protocols.
 - Compute Engine(VM Instance) will run with docker container containing all these services - Container optimized image

EMR Simulator Components - Future



EMR Simulator



EMR Simulator

